ICT2217 Network Security

Lab Exercise: Getting to know the Server

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Access the server via **IPMI (Intelligent Platform Management Interface)** – Out of Band Management (OOB) Port, which is usually also called a **BMC (Baseboard Management Controller)**
- Power on the server through the web-based IPMI
- Access the screen of the server via remote console without any monitors
- Access the Proxmox VM (virtual machine) Hypervisor

Required Hardware

- 1 x Server with IPMI port
- 1 x RJ45 Ethernet Cable
- 1 x Laptop

Required Software

- Web Browser

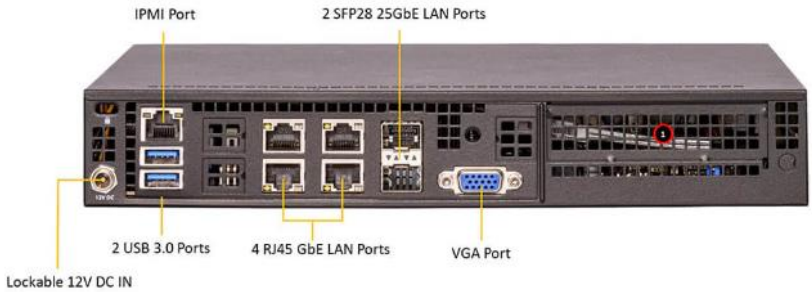
PART 1: Accessing the server via IPMI/OOB Port

1.1 In this lab, you are going to learn the basic management of a server system, specifically the Supermicro Super **Server-Sys-E300-12D-4CN6P-28** series, as shown:



Figure 1.1

1.2 Like a PC or Laptop, this server also has different interfaces for different purposes, even though they look very similar:



	Slot Name	Slot Description
1	Slot 1	PCI-E 4.0 x16 LP

Figure 1.2

1.3 The server can be initially managed using the IPMI/OOB port through an RJ45 Ethernet connection with a pre-configured IP address and embedded web server. This is useful when the server does not yet have an OS (operating system) installed. After the OS is installed, the server can be accessed and managed through SSH remote access.

1.4 Refer to 1.2 Figure, connect the Lab PC or your laptop to the "IPMI Port" using an RJ45 Ethernet cable.

1.5 Once connected, and the status is "up", you can test the connection to the embedded IPMI web server using a PING tool to the *pre-configured* IP address **10.0.0.1**

Note: *You need to statically configure your PC IP to any IP address in the same subnet (i.e., 10.0.0.0/24), for example,* ***10.0.0.100.***

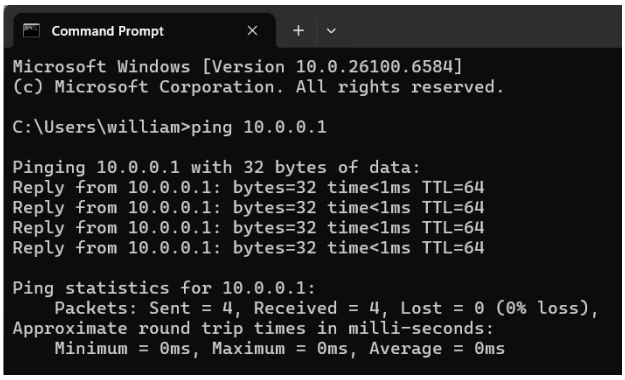


Figure 1.3

1.6 If the ping is successful, it means the embedded IPMI web server is running, and you can test to access the web page using any web browser software with the URL http://10.0.0.1

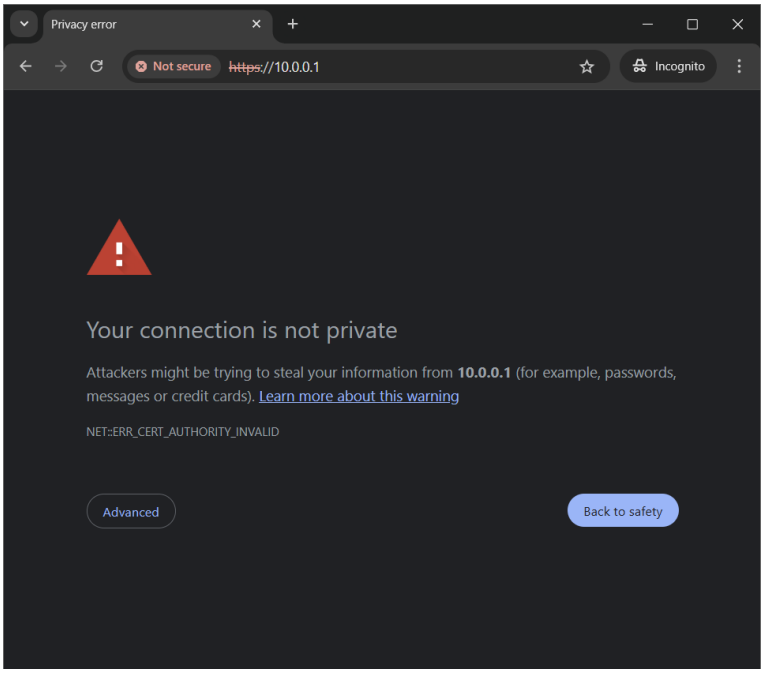


Figure 1.4

1.7 You may see some “Privacy error” due to the invalid secure HTTP connection (HTTPS) certificate. Please ignore the message by clicking the “Advanced” button, and then click the “Proceed to 10.0.0.1 (unsafe)”. It is OK since we know that we are accessing our own server.

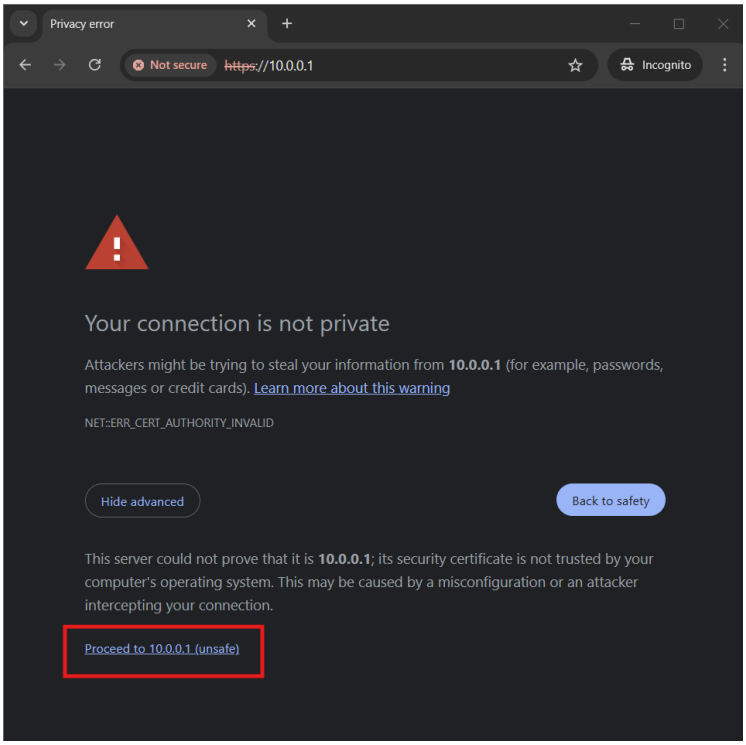


Figure 1.5

1.8 If it is successful, the Supermicro BMC (same as IPMI) Login page will appear, and you need to enter the credentials. Please use the username **sitstudent** and the password as shown on the whiteboard.

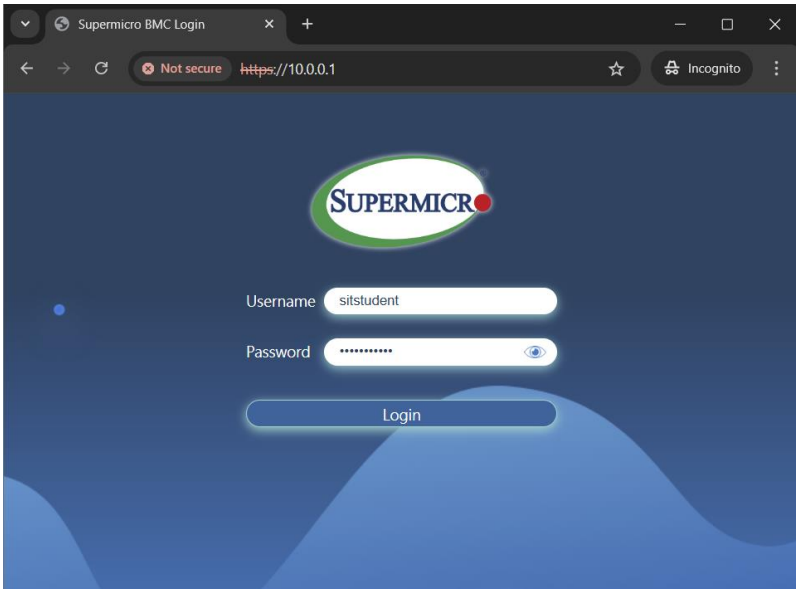


Figure 1.6

1.9 After successful login, the Supermicro BMC will show the dashboard related to the server system's status and configuration. You may check all the options if you want to.

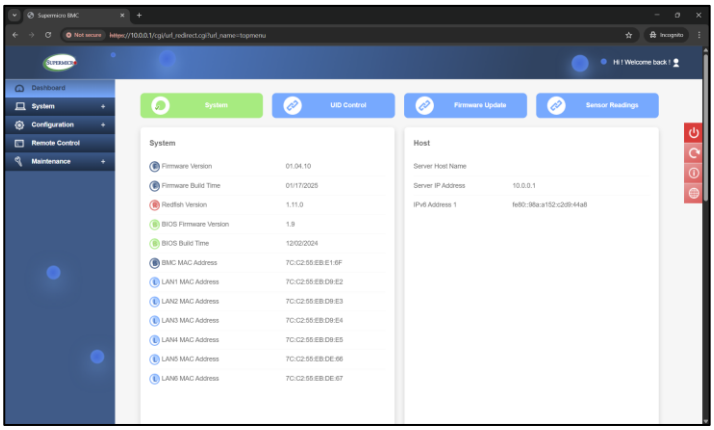


Figure 1.7

PART 2: Power on the server via IPMI Web Console

2.1 By default, the server system will be in a **Power-Off** state when the server power plug is connected to the power outlet. So, it needs a manual **Power-On** through the BMC Dashboard. Please navigate the mouse to the power button in the upper right corner. To power on, please click the button as shown in the picture below.

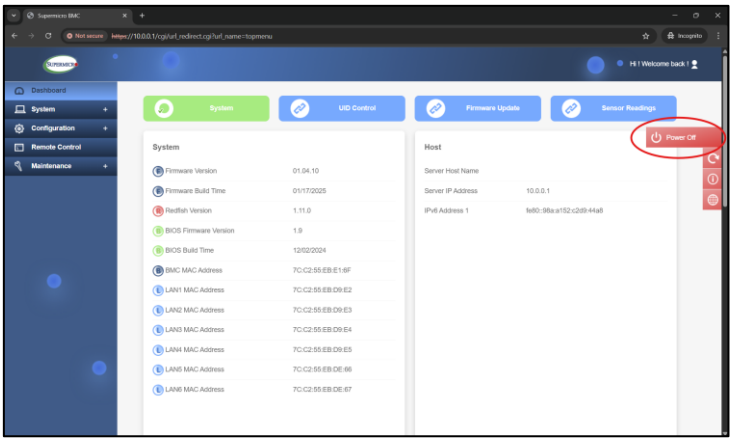


Figure 2.1

2.2 Check or tick the “Power ON” and click “Apply” to confirm the power-on process.

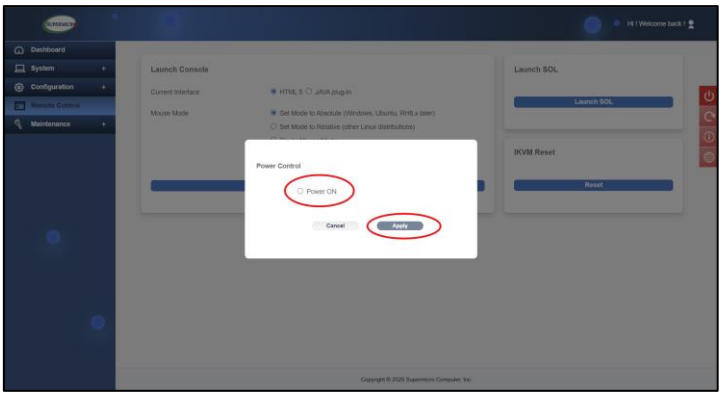


Figure 2.2

2.3 Once it is completed, make sure the page below appears and the power button in the right corner of the page is changed to “Green”, which indicates the server is successfully powered on.

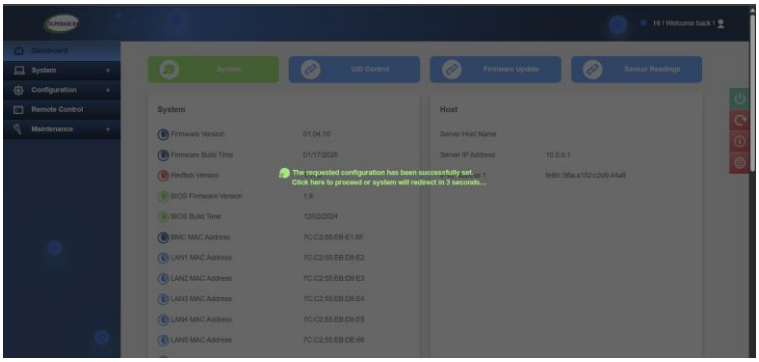


Figure 2.3

PART 3: Accessing the screen of the server via remote console

3.1 It is important to check the server status after it is successfully powered up or to see the progress of the server boot process. Accessing the server's screen via remote console can be done through the "Remote Control" menu on the Supermicro BMC Dashboard.

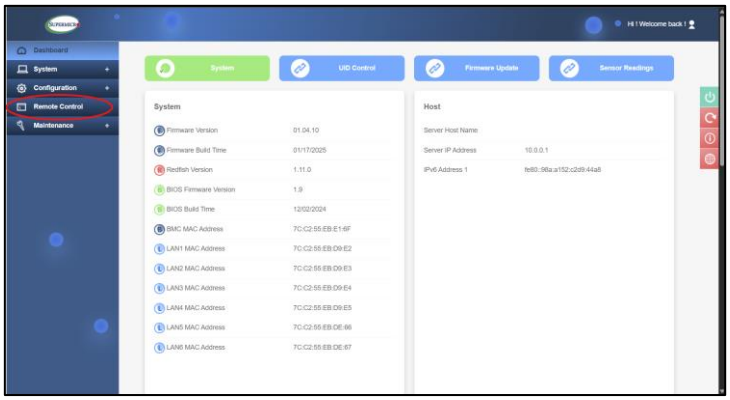


Figure 3.1

3.2 The remote console will also be using a web interface with an optional interface using HTML5 and JAVA plug-in. For simple access, select HTML5 as the "current interface" and click the "Launch Console".

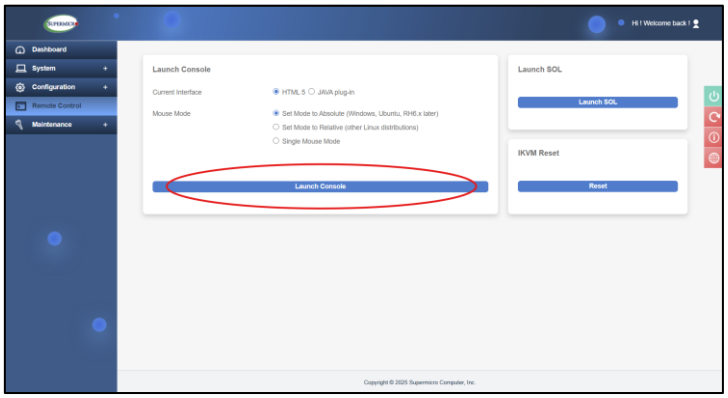


Figure 3.2

3.3 The remote console will be opened on a new page, and it will show the Proxmox Hypervisor Console.

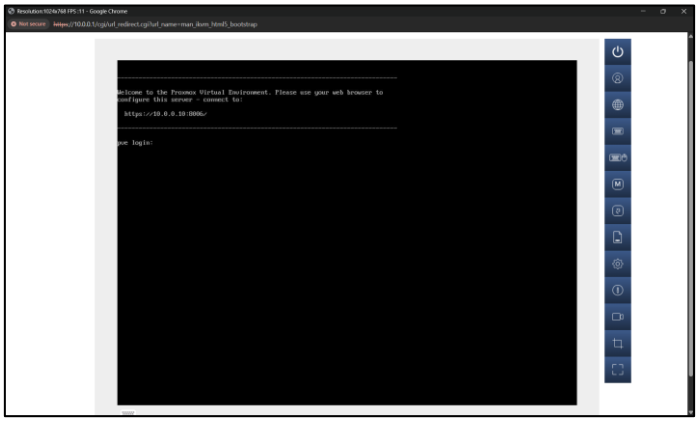


Figure 3.3

PART 4: Checking the Proxmox Hypervisor Management via Web Dashboard

4.1 Proxmox Hypervisor is accessible from the CLI console and the web management dashboard for easy management of virtual machines. To check the access to the Proxmox web dashboard, connect your laptop to port "LAN 1" using an RJ45 Ethernet cable, as shown in Figure 4.1 below.



Figure 4.1

4.2 After it is connected, you can access the Proxmox web management dashboard by opening this URL in your web browser.

https://10.0.0.10:8006

If a privacy error page appears, expand "advanced" and click on "Proceed to 10.0.0.10" to continue.

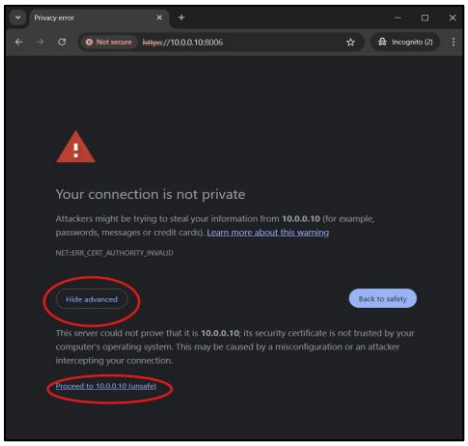


Figure 4.2

4.3 If it is successful, you will see this page and have completed the lab exercise! No need to key in username and password because we will do it in the next lab.

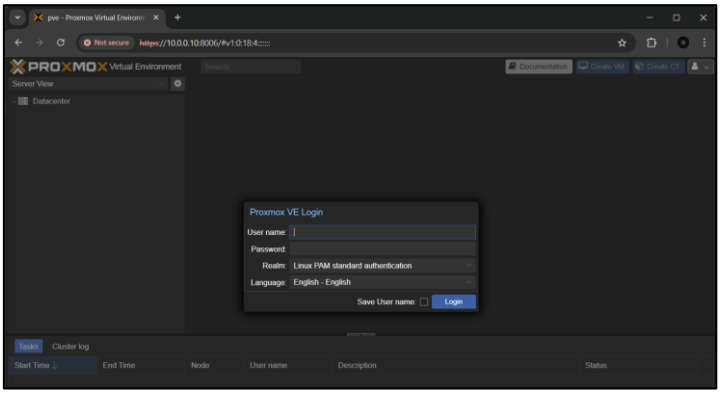


Figure 4.3

PART 5: Powering off the server gracefully

5.1 It is always important to power off the server **gracefully** after it is powered on. This can help prolong the server's lifespan and prevent any OS (operating system) corruption or crashes. There are several ways to gracefully power down the server, either through the BMC Dashboard or through the OS itself (in this case, the Proxmox interface). In this lab, the server will power down through the BMC Dashboard. Please navigate the mouse to the power button in the upper right corner and click the button as shown in the picture below.

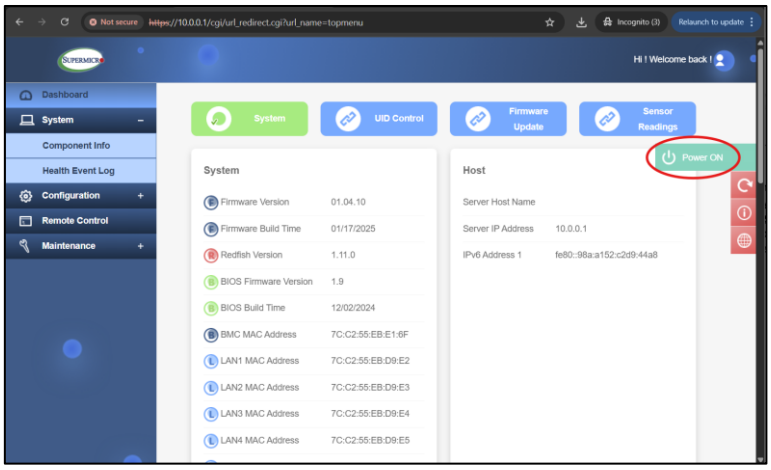


Figure 5.1

5.2 Check or tick the “Graceful Shutdown” and click “Apply” to confirm the power-off process.

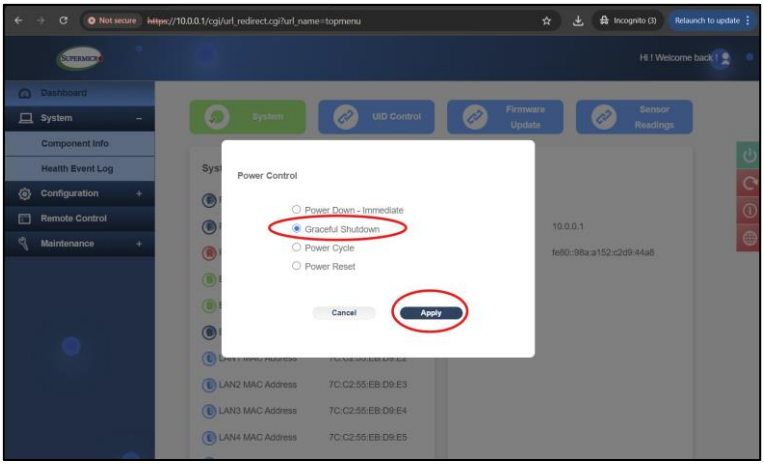


Figure 5.2

5.3 You can verify the server had been powered off by the remote console page or by no longer getting a feel on the breeze produced from the fan outlet located on the slot grill.

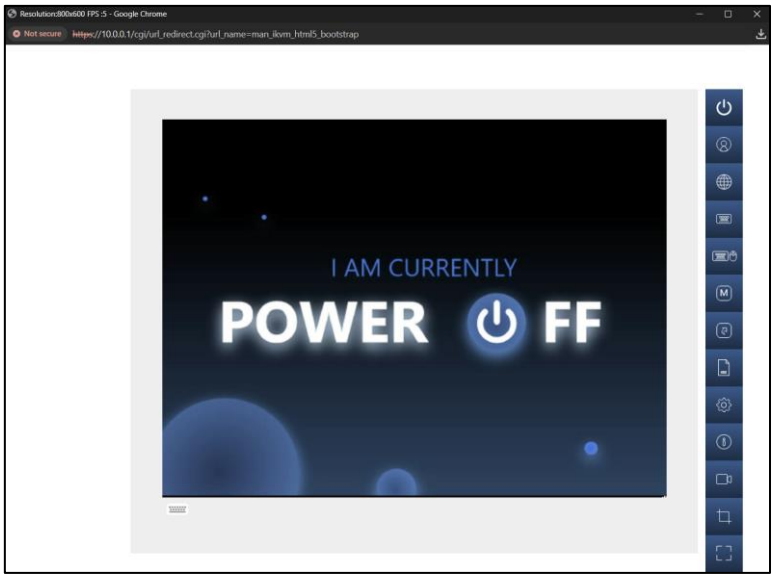


Figure 5.3



Figure 5.4

5.3 Congratulations! You have completed the lab!